

2.8 Ethical and Pedagogical Considerations

LLM assistance in educational settings raises integrity and dependency concerns (Finnie-Ansley et al., 2022). Ask mode’s read-only contract supports formative learning—students can request explanations without automated submission of solutions. Code mode requires manual insertion, preserving a deliberate human step before code enters the notebook. Agentic mode automates edits and should be positioned as an advanced workflow tool rather than a default for graded assignments without instructor policy.

From a safety perspective, constraining tools to six vetted cell operations reduces the attack surface compared with arbitrary code execution or unrestricted web API access (Patil et al., 2023; Schick et al., 2023). Session binding via URL and `tab_id` limits cross-notebook accidents when multiple kernels are open.

2.9 Synthesis: Design Principles for Notebook Agents

Drawing on the surveyed literature, five design principles informed the methodology in Chapter 3:

1. **Ground before generate.** Retrieval-augmented and in-context grounding (Lewis et al., 2020) require fresh notebook serialisation before mutating operations—implemented as mandatory round 0 reads in Agentic mode.
2. **Bound autonomy.** Agentic systems without caps risk runaway tool loops (Yao et al., 2023; Qin et al., 2024); the two-call contract and six-tool surface enforce predictability.
3. **Respect positional semantics.** Notebook cells are not files; bulk edits need deterministic reordering (Rule et al., 2019).
4. **Separate explanation from execution.** Usability research shows users need predictable edit semantics (Vaithilingam et al., 2022; Mozannar et al., 2024). This motivates Ask, Code, and Agentic mode separation.
5. **Evaluate against contracts.** Metrics must match what the system actually does (Hou et al., 2024)—dispatch scoring for Agentic, side-effect-free rubrics for Ask and Code.

These principles bridge the gap between general-purpose LLM tooling and the constraints of closed Kaggle notebook editors identified throughout this chapter.